6. Automation Suite & Workflow Options – Smart Lost and Found Management System

Aim

To build the automation layer of the lost and found application with Zoho Creator form workflows and Deluge scripts, and to study the other workflow options of the platform.

Workflow Mapping Overview

Workflow type	Configuration / purpose	Status
Form workflow	Seven workflows: Validate Lost Item Date, Check_Duplicate_Lost_Item, Validate Found Item Date, Notify Admin of Found Item, Validate Claim Date, Send Claim Email Notification, Update_Found_Item_Status	Created
Schedule	Runs a script at a set time or interval	Studied
Approval	Routes a record to approvers who accept or reject it	Studied
Payment	Collects online payment through a gateway	Studied
Blueprint	Moves a record through defined stages	Studied
Batch workflow	Runs an action on many records that match a criteria	Studied
Report workflow	Adds a custom action to a report	Studied
Function	Reusable custom Deluge function	Studied

Form workflows and Deluge scripts

Four workflows run on validate and can stop a submission with cancel submit; three run on successful submission.

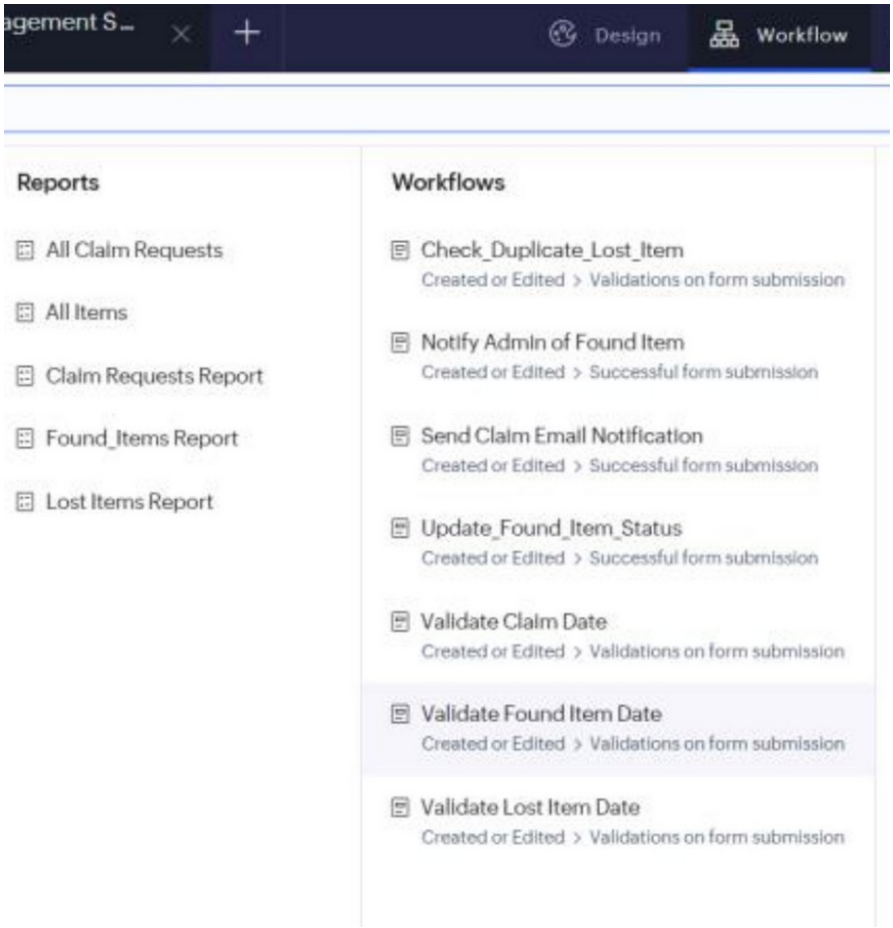


Fig. 6.1: The seven form workflows of the application

DS-02 Duplicate lost report (Lost_Items, On Validate)

```
if(Lost_Items[Item_Name == input.Item_Name
   && Date_Lost == input.Date_Lost
   && Location_Lost == input.Location_Lost].count() > 0)
{
    alert "This lost item has already been reported.";
    cancel submit;
}
```

This check can also stop an edit of an existing lost record that is saved without changes, because the record being edited is counted as a match (BUG-01). Adding the condition ID != input.ID to the criteria is the planned fix.

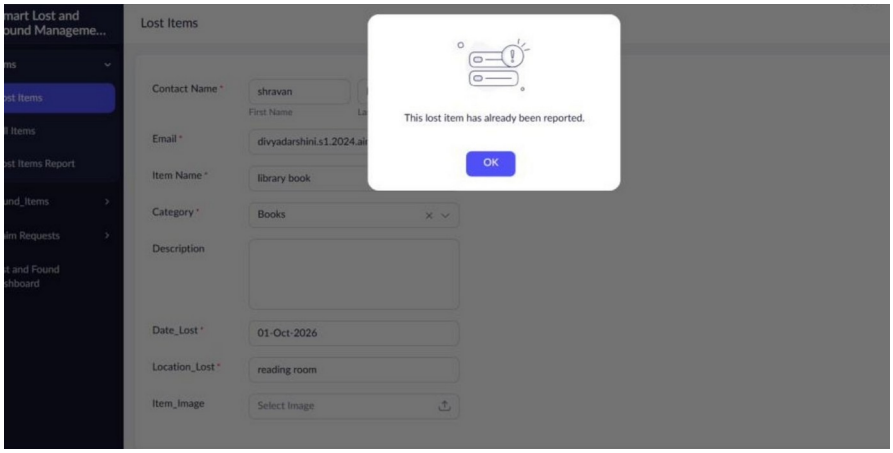


Fig. 6.2: Duplicate lost item message

DS-05 Claim date validation (Claim_Requests, On Validate)

```
if(input.Claim_Date > zoho.currentdate)
{
    alert "Claim Date cannot be in the future.";
    cancel submit;
}
```

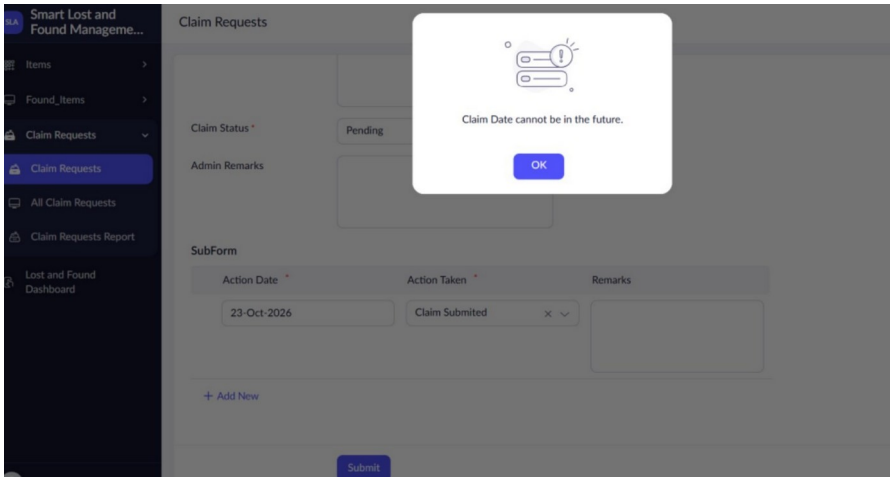


Fig. 6.3: Claim date validation message with the subform visible

DS-06 Email to the administrator (Claim_Requests, On Success)

```
sendmail
[
    from :zoho.adminuserid
    to :"darshanabinav.rk.2024.aiml@rajalakshmi.edu.in"
    subject :"New Claim Request Submitted"
    message :"A new claim request has been submitted. Please review it in
the Smart Lost and Found Management System."
]
```

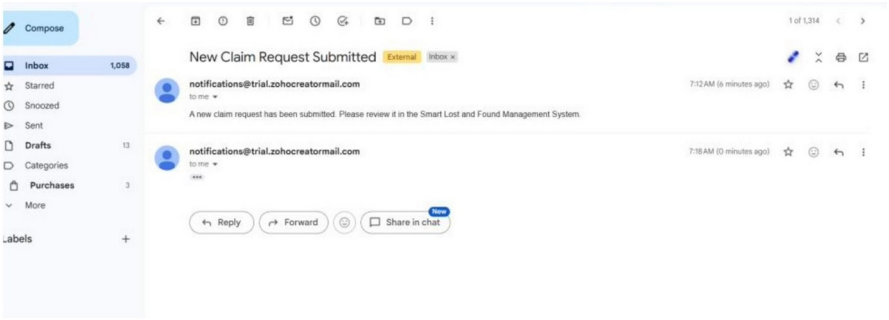
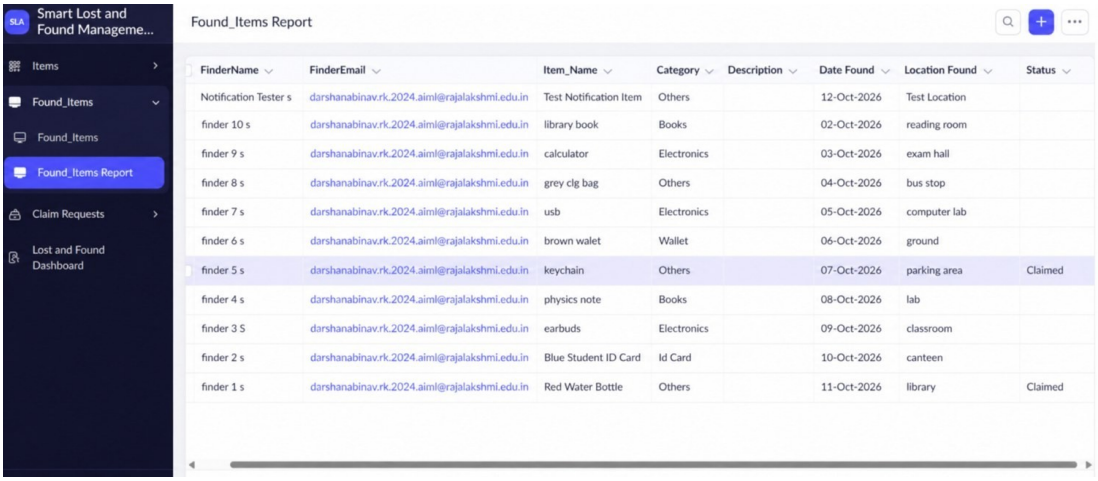


Fig. 6.4: Notification email received in the inbox

DS-07 Mark the found item as Claimed (Claim_Requests, On Success)

```
if(input.Claim_Status == "Approved")
{
    foundRec = Found_Items[ID == input.Found_Items];
    foundRec.Status = "Claimed";
}
```

DS-07 runs after a claim is saved and, when the status is Approved, sets the Status of the linked found item to Claimed. In the sample data this worked for two of the four approved claims (Red Water Bottle and keychain). The found items library book and Blue Student ID Card are not marked Claimed although their claims are Approved; this is recorded as BUG-02 and is still open.



FinderName	FinderEmail	Item_Name	Category	Description	Date Found	Location Found	Status
Notification Tester s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	Test Notification Item	Others		12-Oct-2026	Test Location	
finder 10 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	library book	Books		02-Oct-2026	reading room	
finder 9 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	calculator	Electronics		03-Oct-2026	exam hall	
finder 8 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	grey clg bag	Others		04-Oct-2026	bus stop	
finder 7 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	usb	Electronics		05-Oct-2026	computer lab	
finder 6 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	brown walet	Wallet		06-Oct-2026	ground	
finder 5 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	keychain	Others		07-Oct-2026	parking area	Claimed
finder 4 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	physics note	Books		08-Oct-2026	lab	
finder 3 S	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	earbuds	Electronics		09-Oct-2026	classroom	
finder 2 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	Blue Student ID Card	Id Card		10-Oct-2026	canteen	
finder 1 s	darshanabinav.rk.2024.aiml@rajalakshmi.edu.in	Red Water Bottle	Others		11-Oct-2026	library	Claimed

Fig. 6.5: Found_Items Report showing Claimed for two of the four approved claims

DS-01 and DS-03 reject a lost date and a found date in the future in the same way as DS-05, and DS-04 sends the found-item email in the same way as DS-06.

Other workflow types

The remaining workflow types were studied. Table 6.1 lists what each type does and where it could be used in this application.

Workflow type	What it does	Possible use here
Schedule	Runs a script at a set time or interval	Weekly mail to the administrator with the number of found items still Available

Workflow type	What it does	Possible use here
Approval	Routes a record to approvers who accept or reject it	Sending each claim to the administrator instead of editing Claim Status by hand
Blueprint	Moves a record through defined stages with rules at each step	Claim stages Pending, Under Verification, Approved and Rejected
Batch workflow	Runs an action on many records that match a criteria	Marking found items that stay Available for 30 days for review
Report workflow	Adds a custom action to a report	An Approve action on the Claim Requests Report
Payment	Collects online payment through a gateway such as PayPal or Stripe	Collecting a handling fee, if one were ever charged
Function	Reusable Deluge function called from other scripts	Days an item has stayed unclaimed (see Experiment 14)

Table 6.1: Other workflow types and their possible use

Procedure

1. Open the Workflow tab, choose the form and click New Workflow.
2. Select the trigger (On validate or On successful submission) and write the Deluge script.
3. Save and enable the workflow.
4. Submit test records and check the alert, the email and the status change.

Result

Seven form workflows were created. The claim-date check, the duplicate check and the claim notification email were tested and behaved as designed (Figures 6.2 to 6.4). The lost-date check, the found-date check and the found-item email use the same pattern as the tested ones. The status update worked for two of the four approved claims (BUG-02). The schedule, approval, Blueprint, batch, report, payment and function workflow types were also studied.